

Assignment Category: CAT_10

Requirement Video:  Pet Adoption Requirement.mp4

Pet Adoption Platform

Project Overview and Discussion

You are going to build a full-stack Pet Adoption Platform using the **MERN Stack** along with authentication and secure backend APIs.

The system allows users to explore pets available for adoption (dogs, cats, birds, rabbits, etc.), view detailed pet profiles, and submit adoption requests.

This project reflects a real-world pet adoption portal where users can browse pets, request adoption, and manage their requests, while pet owners or shelters can manage listings and handle adoption requests.

How the System Works

- Users can browse all available pets
- Users can view detailed information about each pet
- Authenticated users can submit adoption requests
- Users can manage their own requests
- Pet owners can add, update, and delete pet listings
- Pet owners can approve or reject adoption requests
- All data will be stored and managed in **MongoDB**

Ensure the Following Things to Get 100% Mark

GitHub Commits

- **Minimum 15 meaningful commits** (client side)
- **Minimum 8 meaningful commits** (server side)

Readme.md

Include:

- Project Name
- Purpose
- Live URL
- Features (at least 5 bullet points)
- NPM Packages Used

Other Rules

- **No Lorem Ipsum text**
- No default alert() for messages
- Use toast/UI-based notifications
- Deploy using **Vercel / Render**
- No error on route reload
- Logged-in user should not redirect on private route reload
- Responsive design (mobile/tablet/desktop)
- Secure **MongoDB** credentials using environment variables
- Design should be clean and recruiter-friendly
- Do not copy any assignment/module project

Deployment Guideline

- Ensure server does not throw CORS / 404 / 504 issues
- Live site should work perfectly
- Reloading any route should not throw error
- Logged-in users must not redirect to login on private route reload

Main Requirements

Layout Structure

- **Navbar / Header**
 - Include: Logo + Website Name, Home, All Pets, My Requests (Private), Add Pet (Private)
 - If user logged in: Profile Dropdown (Dashboard, Logout)
 - If not logged in: Show Login button
- **Footer**
 - Include: Contact Information, Social Links, Copyright
- **Dashboard Layout**
 - My Requests, Add Pet, My Listings

Home Page

- **Banner Section**
 - Eye-catching hero section, Title + short description, “Adopt Now” button
- **Dynamic Section (Required)**
 - Featured Pets Section: Minimum 6 pets, Card info with “View Details” button
- **Extra Static Sections**
 - Why Adopt Pets, Success Stories, Pet Care Tips
 - Add any 2 additional static sections of your choice. You can decide the section names and content based on your creativity.

Authentication

User Login

- **Email**
- **Password**
- Login button
- If success → redirect to desired route/home
- If error → show toast
- Also include:
- Register link
- Google Login

User Registration

- **Name**
- **Email**
- **Photo URL**
- **Password**
- **Confirm Password**
- **Login Link**
- Password Validation
- At least 6 characters
- One uppercase letter
- One lowercase letter
- Password & Confirm Password must be same
- If success → redirect to login/home
- If error → show message

Do NOT implement email verification or forget password!!!

CRUD Operations

All Pets (Main Layout) (Public Route)

Display all pets in cards with a "View Details" button (Navigates to the Pet's details page)

!!! If user clicks "Adopt Now" without login → redirect to login

Add Pet (Dashboard Layout) (Private Route)

Fields:

- Pet Name
- Species (Dog/Cat/Bird/etc.)
- Breed
- Age
- Gender
- Image URL (imgbb/postimage)
- Health Status
- Vaccination Status
- Location
- Adoption Fee
- Description
- Owner Email (auto-filled read only)

If success → Show success toast message and redirect to (My Listings page)

If error → show relevant message

Store in: MongoDB pets collection

Pet Details + Adoption (Main Layout) (Private Route)

Display full pet details along with an Adoption Form.

Implement the form in a side panel, right section, or a modal:

- Pet Name (Read Only)
- User Name (Read Only)
- User Email (Read Only)
- Pickup Date
- Message
- Adopt Button

Default status: "pending"

My Listings (Dashboard Layout) (Private Route)

Display:

- Title
- Stats (Total Listings, Available, Adopted)
- Show Pet's card:
 - ★ Image
 - ★ Name
 - ★ Price
 - ★ Requests Button (Open Requests Modal)
 - ★ Edit Button (Open Update Pet Modal/Navigate Update Pet Page)
 - ★ View Button (Navigate to Pet details page)
 - ★ Delete Button

Requests Modal:

Display:

- Title
- Requested User Name, Email
- Pickup Date
- Approve Button
- Reject Button

If already approved or rejected → don't show Approve Reject Button.

Update (Modal/Page) / Delete Pet (Modal) (Private Route)

Only pet owner can:

- Update pet form (Same as Add Pet Form)
- Delete pet (with confirmation)

My Requests (Private Route)

Display:

- **Pet Name**
- Request Date
- Pickup Date
- Status (Pending/Approved/Rejected)

- View Button (Navigate to Pet Details Page)
- Cancel Button (Delete the Request)

Other Requirements

Loading Page

Show spinner while fetching data

Not Found Page

- Custom 404 page
- Friendly message
- Back to Home button

Challenges

Search, Filter & Sorting In All Pets Page

Implement advanced pet browsing features in the **All Pets Page**:

- Search pets by name
- Filter pets by species

Use **MongoDB** operators:

- \$regex
- \$in

Follow this documentation: [\\$regex-Documentation](#), [\\$in-Documentation](#)

Adoption Control

- Pet owners are not allowed to submit adoption requests.

- If multiple adoption requests exist, only one can be accepted from the Requests Modal
- When a request is approved:
 - Mark the pet as adopted
 - Prevent further requests

JWT Authentication

- Generate token
- Store in **HTTPOnly cookies**
- Verify in middleware
- Protect private routes

Optional Requirements (Choose Any 2)

- Dark/Light Theme Toggle
- Framer Motion Animation
- Wishlist Feature

What to Submit

1. Client-side GitHub Repository
2. Server-side GitHub Repository
3. Live Website Link